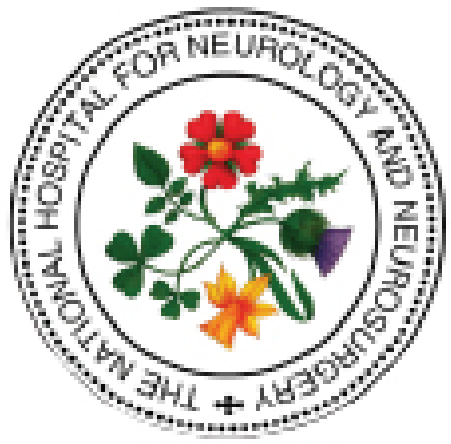




TARGETED TEMPERATURE MANAGEMENT (TTM) in ACUTE BRAIN INJURY

Targeted temperature management (TTM) is the process of controlling core body temperature to maintain **normal body temperature** (TTMnorm)

Neurogenic fever is associated with poor outcomes amongst patients with intracerebral haemorrhage (ICH), subarachnoid haemorrhage (SAH), or acute ischaemic stroke (AIS) and should be treated - *fever can increase the brain's metabolic demand, exacerbating ischaemic injury, leading to cerebral vessel vasodilation and worsening intracranial pressure (ICP)*

INDICATIONS FOR USE

Neurogenic fever is associated with poor outcomes amongst patients with ICH, SAH, or AIS and should be treated. Fever can increase the brain's metabolic demand, exacerbating ischaemic injury and leading to cerebral vessel vasodilation, ultimately worsening intracranial pressure (ICP).

- TTM should be used **reactively** in patients with ICH, SAH, or AIS in response to neurogenic fever (*non-infectious fever*)
- TTM should be **initiated** if patient temperature increases to **37.5°C** and infection is excluded
- TTM should be initiated as **rapidly** as possible once fever detected if pharmacological treatment has not controlled temperature **within 1h** of administration - *1st line paracetamol / 1g/qds*

There is limited evidence to support prophylactic TTM after ICH, SAH, or AIS regardless of core temperature and it carries additional risks such as masking fever associated with infection and should be avoided

COLOMBIA BEDSIDE SHIVERING ASSESSMENT SCALE		
SCORE	TYPE OF SHIVERING	LOCATION
0	NONE	No shivering detected on palpation of masseter, neck or chest muscles
1	MILD	Shivering localised to neck and thorax only
2	MODERATE	Shivering involves gross movement of upper extremities (in addition to neck and thorax)
3	SEVERE	Shivering involves gross movements of trunk and upper and lower extremities

ANTI-SHIVERING PROTOCOL*			
STEP		INTERVENTION	DOSE
0	BASELINE	SKIN COUNTER WARMING	32-38C
1	MILD SEDATION	OPIOID	Starting dose 25mcg/h
2	MODERATE SEDATION	OPIOID +	
3	DEEP SEDATION	PROPOFOL	50-75mcg/kg/min
4	NEUROMUSCULAR BLOCKADE	VECURONIUM	0.1mg/kg IV

References

- P. J. D. Andrews, V. Verma, M. Healy, A. Lavinio, C. Curtis, U. Reddy, J. Andrzejowski, A. Foulkes and S. Canestrini: Updated consensus recommendations. Targeted temperature management in patients with ICH, SAH, or AIS.; British Journal of Anaesthesia 2023
- Doyle JF, Schortgen F. Should we treat pyrexia? And how do we do it? Crit Care 2016; 20: 303
- Rincon F, Hunter K, Schorr C, Dellinger RP, Zanotti-Cavazzoni S. The epidemiology of spontaneous fever and hypothermia on admission of brain injury patients to intensive care units: a multicenter cohort study J Neurosurg 2014; 121: 950e60 774

TARGETED TEMPERATURE MANAGEMENT in SEDATED PATIENTS if TEMPERATURE $\geq 37.5^{\circ}\text{C}$

WHEN TO USE TTM

- Core temperature should be measured **continuously** or at minimum **1 hourly** to enable effective identification, treatment, and monitoring of neurogenic fever
- TTM should be initiated as **rapidly** as possible once fever detected if pharmacological treatment has not controlled temperature **within 1h** of administration - *1st line paracetamol / 1g/qds*

The **TARGET TEMPERATURE** for patients with ICH, SAH, or AIS who develop neurogenic fever is **37.0C $\pm$ 0.5C**

Choose **NORMOTHERMIC** setting on Arctic Sun to prevent risk of hypothermia

SHIVERING DURING TTM

Shivering should be managed with non-pharmacological and pharmacological methods

- It can cause cerebral and metabolic stress, and negate beneficial effects of TTM
- It should be treated in a stepwise approach using non-sedating interventions, such as counter-warming, before use of sedatives or neuromuscular block and using a bedside shivering assessment scale, to help characterise shivering and support decision-making

COUNTER-WARMING

Counter-warming of skin on head, hands or arms (*using gloves, hats, socks, towels and blankets*) can reduce shivering without pharmacological intervention, whilst allowing core temperature to be controlled

Apply **Bair Hugger** for surface warming

PHARMACOLOGICAL MANAGEMENT

Decisions guided by level of consciousness or risks of pharmacological interventions (e.g. bradycardia / hypotension)

AWAKE PATIENTS

Paracetamol
Clonidine

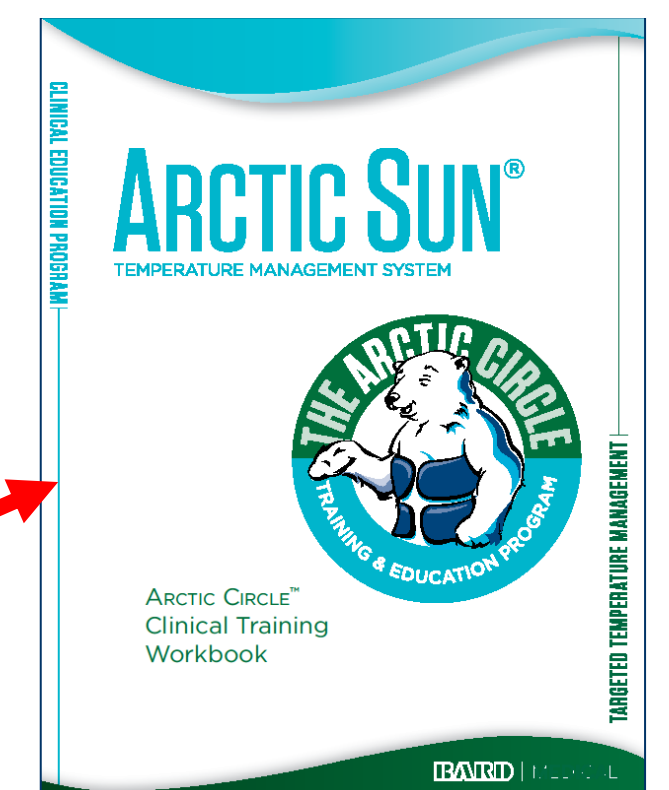
SEDATED PATIENTS

Fentanyl
Propofol
Midazolam
Neuromuscular paralysis

DURATION OF TTM

- TTM should be maintained for as long as there is potential for secondary brain damage
- Controlled rewarming should be at a rate of **<1°C per day**

Arctic Sun TTM
Non-invasive device modulating patient temperature by circulating chilled water in pads directly adhered to skin



CONSENSUS GUIDELINES

